# Configuring Local SSH Access to Provisioned Linux Containers

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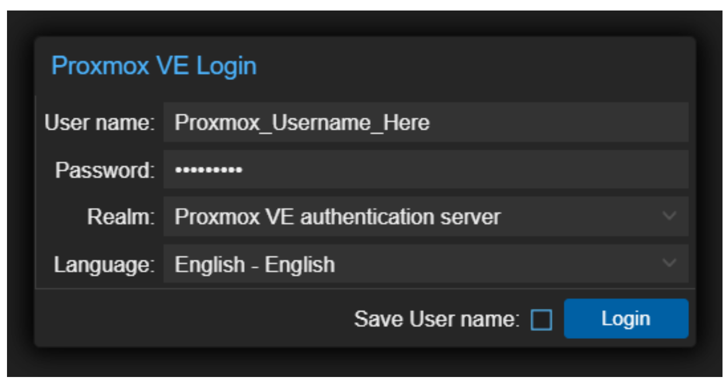
This guide outlines the process of creating a Linux user account within a Proxmox-hosted Linux container to enable local SSH access. It assumes that the required Proxmox credentials and target container information have already been provided.

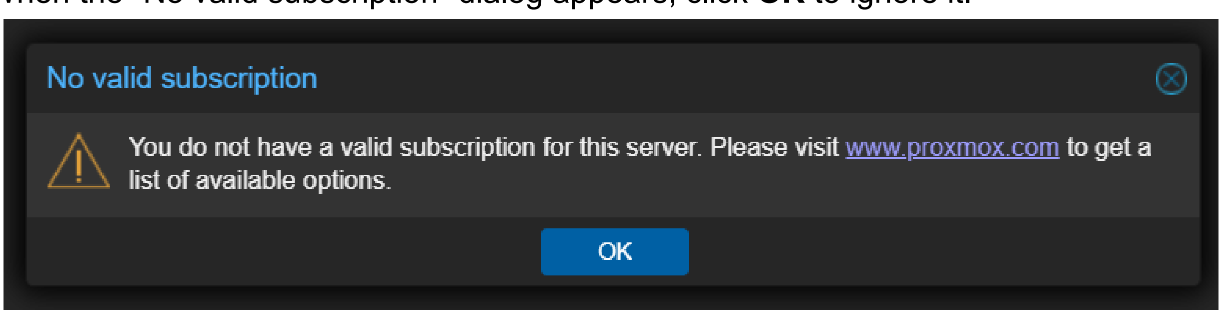
- **Proxmox** is an open-source virtualization platform used to manage virtual machines and Linux containers on a centralized infrastructure.
- **Linux container (LXC)** is a lightweight, isolated operating system environment that shares the host system's kernel while maintaining its own users, processes, and file system.
- **Secure Shell (SSH)** is a network protocol that enables secure remote access to a Linux system through a command-line interface.
    - In this guide, local SSH access refers to connecting directly to a Linux container from a user's local machine using an SSH client, allowing the user to securely access and manage the container through the command line without using the Proxmox web interface console.

## Log in to Proxmox

1. Go to [https://ccscloud.dlsu.edu.ph/](https://ccscloud.dlsu.edu.ph/)
    - Click proceed/continue when prompted



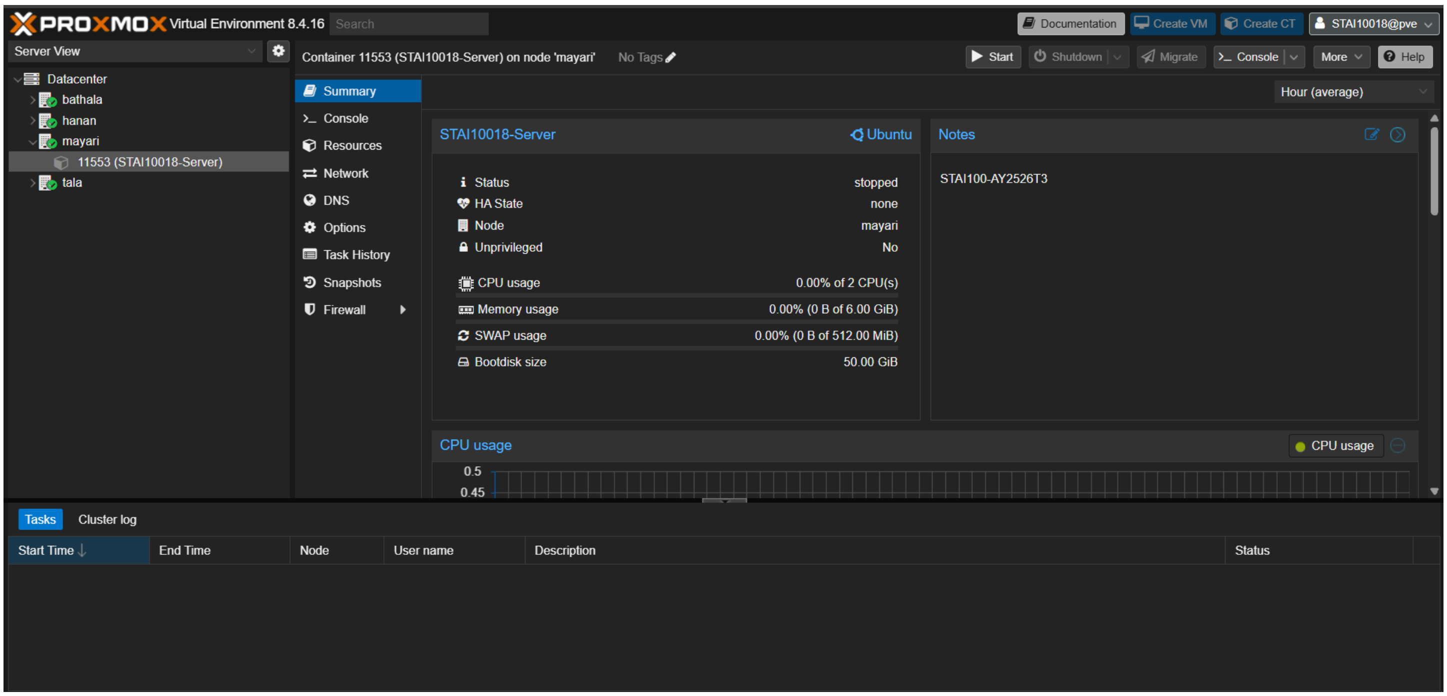

2. Fill in the appropriate fields with your provisioned credentials:
    - `Proxmox Username`
    - `Proxmox Password`
    - `Proxmox Realm`
3. Click **Login**.
4. When the “No valid subscription” dialog appears, click **OK** to ignore it.



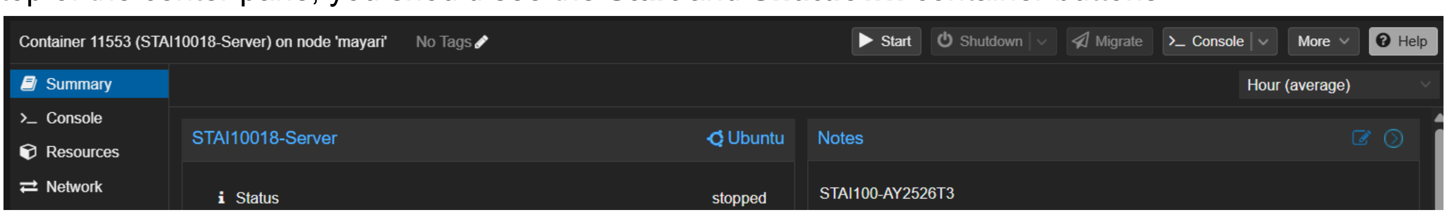
## View Linux Container Information

In the top-right corner, you should see the dropdown menu toggle for account-related settings.


On the left sidebar, you'll see the different nodes within CCS Cloud. If you click the `Proxmox Node`, you should see your `LXC ID` and `LXC Name` (e.g., **mayari** and **11553 (STAI10018-server)** in the screenshot).



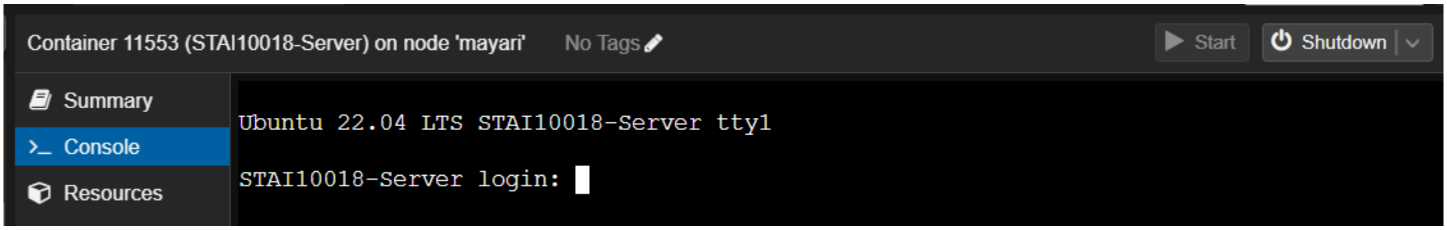

Clicking the container will display its summary information in the center pane. Newly created Linux containers are powered off by default, so the container status will initially appear as **stopped**. At the top of the center pane, you should see the **Start** and **Shutdown** container buttons.




## Creating a New Linux User

Local SSH access is not permitted using the provided **root** account. To enable SSH access, create a new Linux user account through the Proxmox web interface console. If the Linux container is not already running, **start** it first before proceeding with the user creation steps.

1. Open the **Console**. The Console button in the top bar opens a new pop-out window, while the Console tab in the center pane (below the Summary tab) opens the console within the main interface.




2. Input the LCX credentials. Note that all the password prompts will not display any visual feedback while you type or paste the password.

```
Ubuntu 22.04 LTS STAI10018-Server tty1

STAI10018-Server login: LCX_Root_Username
Password: LCX_Root_Password
```

3. On successful login, you should see the Ubuntu welcome message.

```
Welcome to Ubuntu 22.04 LTS (GNU/Linux 6.8.12-18-pve x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

root@STAI10018-Server:~#
```

4. Create a new Linux user by running **`adduser your_new_username`**. You will be prompted to set a password for the new account. Afterward, the system will request additional user information, which can be left blank.

```
root@STAI10018-Server:~# adduser kmdk
Adding user `kmdk' ...
Adding new group `kmdk' (1000) ...
Adding new user `kmdk' (1000) with group `kmdk' ...
Creating home directory `/home/kmdk' ...
Copying files from `/etc/skel' ...
New password: Non_Root_Password
Retype new password: Non_Root_Password
passwd: password updated successfully
Changing the user information for kmdk
Enter the new value, or press ENTER for the default
```

```
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y
```

5. Grant sudo privileges to the new account by running **`usermod -aG sudo your_new_username`**

```
root@STAI10018-Server:~# usermod -aG sudo kmdk
```

6. Run **`exit`** to log out of the current session. Then, log back in using the new user credentials to confirm that the account was created successfully. Note the change in the console prompt, which indicates you are now logged in as the new user rather than root.

```
root@STAI10018-Server:~# exit
logout

Ubuntu 22.04 LTS STAI10018-Server tty1

STAI10018-Server login: kmdk
Password: Non_Root_Password
Welcome to Ubuntu 22.04 LTS (GNU/Linux 6.8.12-18-pve x86_64)
  [...]

kmdk@STAI10018-Server:~$
```

## Creating a New Linux User

You can begin deploying applications in the Linux container using the Proxmox web interface console. If you prefer to work without logging into Proxmox, you can configure local SSH access instead. To know if your terminal has an SSH client installed, run **`ssh -V`**

1. In your local terminal, run the following command:
   - **`ssh -p External_Port_22 Non_Root_Username@Public_IP_Address`**

```
~>$ ssh -p 2137 kmdk@103.231.240.130
```

2. On initial SSH connection, you will be prompted to confirm the host key. Enter **yes** to proceed. You will then be prompted for the password. A successful login will display the Ubuntu welcome message. Note the change in the console prompt, which shows that you are now logged in via SSH from your local machine and working directly in the Linux container.

```
~>$ ssh -p 2137 kmdk@103.231.240.130
The authenticity of host '[103.231.240.130]:2137 ([103.231.240.130]:2137)'
```

```
can't be established.
ED25519 key fingerprint is [REDACTED].
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[103.231.240.130]:2137' (ED25519) to the list of
known hosts.
kmdk@103.231.240.130's password: Non_Root_Password
Welcome to Ubuntu 22.04 LTS (GNU/Linux 6.8.12-18-pve x86_64)
  [...]

kmdk@STAI10018-Server:~$
```

3. Close the SSH connection by running **`exit`**. Note the switch back to the console prompt, which indicates you are no longer connected via SSH.

```
kmdk@STAI10018-Server:~$ exit
logout
Connection to 103.231.240.130 closed.

~>$
```

4. Subsequent SSH connections will only prompt for the password.

```
~>$ ssh -p 2137 kmdk@103.231.240.130
kmdk@103.231.240.130's password: Non_Root_Password
Welcome to Ubuntu 22.04 LTS (GNU/Linux 6.8.12-18-pve x86_64)
  [...]

kmdk@STAI10018-Server:~$
```

## Acknowledgments / Disclosure

The following Generative AI tools were used to assist in the making of this document:

- ChatGPT was mainly used to streamline and/or construct sentences. The instructor validated the AI-generated output and modified it as needed.